

EXP 28 – Medical Diagnosis (Prolog)

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Aim

To write a Prolog program that diagnoses common diseases based on symptoms provided by the user by matching them with predefined symptom–disease rules.

Algorithm

1. Accept a list of symptoms from the user.
2. Check whether each required symptom for a disease is present using member/2.
3. Define rules for each disease:
 - Cold → cough, sneezing, runny_nose
 - Flu → fever, body_ache, fatigue
 - Malaria → fever, chills, sweating
4. If all symptoms of a disease match, conclude that disease.
5. Return the diagnosis.

Source Code

prolog

has_symptom(Symptoms, S) :- member(S, Symptoms).

diagnose(Symptoms, cold) :-

has_symptom(Symptoms, cough),
has_symptom(Symptoms, sneezing),
has_symptom(Symptoms, runny_nose).

diagnose(Symptoms, flu) :-

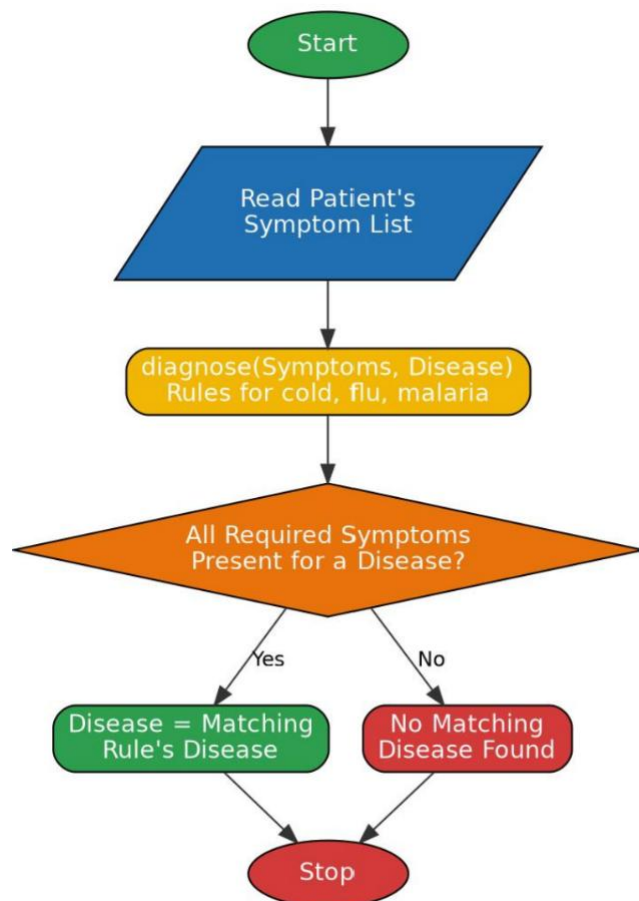
has_symptom(Symptoms, fever),

has_symptom(Symptoms, body_ache),
has_symptom(Symptoms, fatigue).

diagnose(Symptoms, malaria) :-

has_symptom(Symptoms, fever),
has_symptom(Symptoms, chills),
has_symptom(Symptoms, sweating).

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [medical].
true.

?- diagnose([cough, sneezing, runny_nose], Disease).
Disease = cold .

?- diagnose([fever, body_ache, fatigue], Disease).
Disease = flu .

?- diagnose([fever, chills, sweating], Disease).
Disease = malaria .
```

Result

The program correctly identifies diseases by checking whether the user-provided symptoms match the predefined symptom patterns for cold, flu, or malaria.